

/*

RACE_DAY DATABASE

SQL Server / SSMS

Section C - SQL Database Script

*/

IF DB_ID('Race_Day') IS NULL

BEGIN

CREATE DATABASE Race_Day;

END

GO

USE Race_Day;

GO

-- DROP TABLES

IF OBJECT_ID('dbo.RACE_RESULT', 'U') IS NOT NULL

DROP TABLE dbo.RACE_RESULT;

IF OBJECT_ID('dbo.PAYMENT', 'U') IS NOT NULL

DROP TABLE dbo.PAYMENT;

IF OBJECT_ID('dbo.REGISTRATION', 'U') IS NOT NULL

DROP TABLE dbo.REGISTRATION;

```
IF OBJECT_ID('dbo.RACE_SPONSOR', 'U') IS NOT NULL
    DROP TABLE dbo.RACE_SPONSOR;
```

```
IF OBJECT_ID('dbo.RACE_CATEGORY', 'U') IS NOT NULL
    DROP TABLE dbo.RACE_CATEGORY;
```

```
IF OBJECT_ID('dbo.RACE', 'U') IS NOT NULL
    DROP TABLE dbo.RACE;
```

```
IF OBJECT_ID('dbo.SPONSOR', 'U') IS NOT NULL
    DROP TABLE dbo.SPONSOR;
```

```
IF OBJECT_ID('dbo.LOCATION', 'U') IS NOT NULL
    DROP TABLE dbo.LOCATION;
```

```
IF OBJECT_ID('dbo.ORGANIZER', 'U') IS NOT NULL
    DROP TABLE dbo.ORGANIZER;
```

```
IF OBJECT_ID('dbo.RUNNER', 'U') IS NOT NULL
    DROP TABLE dbo.RUNNER;
```

```
GO
```

```
-- 1. RUNNER
```

```
CREATE TABLE RUNNER
```

```
(  
    runner_id INT IDENTITY(1,1) NOT NULL,  
    first_name NVARCHAR(50) NOT NULL,  
    last_name NVARCHAR(50) NOT NULL,  
    date_of_birth DATE NOT NULL,  
    gender NVARCHAR(10) NOT NULL,  
    email NVARCHAR(100) NOT NULL,  
    phone NVARCHAR(20) NOT NULL,  
    address NVARCHAR(200) NOT NULL,  
  
    registration_date DATE NOT NULL  
        CONSTRAINT DF_RUNNER_registration_date  
        DEFAULT CONVERT(DATE, GETDATE()),  
  
    CONSTRAINT PK_RUNNER  
        PRIMARY KEY (runner_id),  
  
    CONSTRAINT UQ_RUNNER_email  
        UNIQUE (email),  
  
    CONSTRAINT CK_RUNNER_gender  
        CHECK (gender IN ('Male', 'Female', 'Other'))  
);  
  
GO
```

-- 2. ORGANIZER

CREATE TABLE ORGANIZER

(
 organizer_id INT IDENTITY(1,1) NOT NULL,
 organizer_name NVARCHAR(100) NOT NULL,
 contact_name NVARCHAR(100) NOT NULL,
 email NVARCHAR(100) NOT NULL,
 phone NVARCHAR(20) NOT NULL,
 website NVARCHAR(200) NULL,

CONSTRAINT PK_ORGANIZER

 PRIMARY KEY (organizer_id),

CONSTRAINT UQ_ORGANIZER_email

 UNIQUE (email)

);

GO

-- 3. LOCATION

CREATE TABLE LOCATION

(
 location_id INT IDENTITY(1,1) NOT NULL,
 location_name NVARCHAR(100) NOT NULL,

```
address NVARCHAR(200) NOT NULL,  
city NVARCHAR(80) NOT NULL,  
province NVARCHAR(80) NOT NULL,  
postal_code NVARCHAR(10) NOT NULL,  
latitude DECIMAL(9,6) NULL,  
longitude DECIMAL(9,6) NULL,
```

```
CONSTRAINT PK_LOCATION
```

```
    PRIMARY KEY (location_id),
```

```
CONSTRAINT UQ_LOCATION_name_address
```

```
    UNIQUE (location_name, address),
```

```
CONSTRAINT CK_LOCATION_latitude
```

```
    CHECK (latitude IS NULL OR latitude BETWEEN -90 AND 90),
```

```
CONSTRAINT CK_LOCATION_longitude
```

```
    CHECK (longitude IS NULL OR longitude BETWEEN -180 AND 180)
```

```
);
```

```
GO
```

```
-- 4. RACE
```

```
CREATE TABLE RACE
```

```
(
```

race_id INT IDENTITY(1,1) NOT NULL,
race_name NVARCHAR(150) NOT NULL,
description NVARCHAR(500) NOT NULL,
race_date DATE NOT NULL,
location_id INT NOT NULL,
race_type NVARCHAR(10) NOT NULL,
entry_fee DECIMAL(10,2) NOT NULL,
capacity INT NOT NULL,
organizer_id INT NOT NULL,

status NVARCHAR(20) NOT NULL

CONSTRAINT DF_RACE_status

DEFAULT 'Scheduled',

CONSTRAINT PK_RACE

PRIMARY KEY (race_id),

CONSTRAINT UQ_RACE_name_date

UNIQUE (race_name, race_date),

CONSTRAINT CK_RACE_type

CHECK (race_type IN ('5km', '10km', '21km', '42km', 'Other')),

CONSTRAINT CK_RACE_entry_fee

CHECK (entry_fee >= 0),

CONSTRAINT CK_RACE_capacity

CHECK (capacity > 0),

CONSTRAINT CK_RACE_status

CHECK

(

status IN

(

'Scheduled',

'In Progress',

'Completed',

'Cancelled'

)

),

CONSTRAINT FK_RACE_LOCATION

FOREIGN KEY (location_id)

REFERENCES LOCATION(location_id),

CONSTRAINT FK_RACE_ORGANIZER

FOREIGN KEY (organizer_id)

REFERENCES ORGANIZER(organizer_id)

);

GO

-- 5. RACE_CATEGORY

CREATE TABLE RACE_CATEGORY

(
 category_id INT IDENTITY(1,1) NOT NULL,
 race_id INT NOT NULL,
 category_name NVARCHAR(50) NOT NULL,
 gender_eligibility NVARCHAR(10) NOT NULL,
 age_from INT NOT NULL,
 age_to INT NOT NULL,

CONSTRAINT PK_RACE_CATEGORY

 PRIMARY KEY (category_id),

CONSTRAINT UQ_RACE_CATEGORY_name

 UNIQUE (race_id, category_name),

CONSTRAINT CK_RACE_CATEGORY_gender

 CHECK

 (
 gender_eligibility IN
 ('Male', 'Female', 'Open')
),

CONSTRAINT CK_RACE_CATEGORY_age_from

 CHECK (age_from >= 0),


```
CONSTRAINT CK_RACE_CATEGORY_age_to  
    CHECK (age_to >= age_from),
```

```
CONSTRAINT FK_RACE_CATEGORY_RACE  
    FOREIGN KEY (race_id)  
    REFERENCES RACE(race_id)
```

```
);
```

```
GO
```

```
-- 6. REGISTRATION
```

```
CREATE TABLE REGISTRATION
```

```
(  
    registration_id INT IDENTITY(1,1) NOT NULL,  
    runner_id INT NOT NULL,  
    race_id INT NOT NULL,
```

```
    registration_date DATE NOT NULL
```

```
    CONSTRAINT DF_REGISTRATION_registration_date  
    DEFAULT CONVERT(DATE, GETDATE()),
```

```
    status NVARCHAR(20) NOT NULL
```

```
    CONSTRAINT DF_REGISTRATION_status
```

DEFAULT 'Pending',

amount_paid DECIMAL(10,2) NOT NULL

CONSTRAINT DF_REGISTRATION_amount_paid

DEFAULT 0.00,

payment_status NVARCHAR(20) NOT NULL

CONSTRAINT DF_REGISTRATION_payment_status

DEFAULT 'Unpaid',

bib_number INT NULL,

CONSTRAINT PK_REGISTRATION

PRIMARY KEY (registration_id),

CONSTRAINT UQ_REGISTRATION_runner_race

UNIQUE (runner_id, race_id),

CONSTRAINT UQ_REGISTRATION_bib

UNIQUE (race_id, bib_number),

CONSTRAINT CK_REGISTRATION_status

CHECK

(

status IN

('Pending', 'Confirmed', 'Cancelled')

),

CONSTRAINT CK_REGISTRATION_payment_status

CHECK

(

payment_status IN

('Unpaid', 'Paid', 'Refunded')

),

CONSTRAINT CK_REGISTRATION_amount_paid

CHECK (amount_paid >= 0),

CONSTRAINT CK_REGISTRATION_bib

CHECK (bib_number IS NULL OR bib_number > 0),

CONSTRAINT FK_REGISTRATION_RUNNER

FOREIGN KEY (runner_id)

REFERENCES RUNNER(runner_id),

CONSTRAINT FK_REGISTRATION_RACE

FOREIGN KEY (race_id)

REFERENCES RACE(race_id)

);

GO

-- 7. PAYMENT

CREATE TABLE PAYMENT

(

payment_id INT IDENTITY(1,1) NOT NULL,

registration_id INT NOT NULL,

payment_date DATETIME2 NOT NULL

CONSTRAINT DF_PAYMENT_payment_date

DEFAULT SYSDATETIME(),

amount DECIMAL(10,2) NOT NULL,

payment_method NVARCHAR(20) NOT NULL,

transaction_reference NVARCHAR(100) NOT NULL,

status NVARCHAR(20) NOT NULL

CONSTRAINT DF_PAYMENT_status

DEFAULT 'Success',

CONSTRAINT PK_PAYMENT

PRIMARY KEY (payment_id),

CONSTRAINT UQ_PAYMENT_transaction_reference

UNIQUE (transaction_reference),

CONSTRAINT CK_PAYMENT_amount

CHECK (amount > 0),

CONSTRAINT CK_PAYMENT_method

CHECK

(

payment_method IN

('Card', 'EFT', 'Cash', 'Other')

),

CONSTRAINT CK_PAYMENT_status

CHECK

(

status IN

('Success', 'Failed', 'Refunded')

),

CONSTRAINT FK_PAYMENT_REGISTRATION

FOREIGN KEY (registration_id)

REFERENCES REGISTRATION(registration_id)

);

GO

-- 8. RACE_RESULT

CREATE TABLE RACE_RESULT

```
(
    result_id INT IDENTITY(1,1) NOT NULL,
    registration_id INT NOT NULL,
    finish_time TIME(0) NULL,
    overall_position INT NULL,
    category_position INT NULL,
    chip_time TIME(0) NULL,

    status NVARCHAR(10) NOT NULL
        CONSTRAINT DF_RACE_RESULT_status
        DEFAULT 'Finished',

    CONSTRAINT PK_RACE_RESULT
        PRIMARY KEY (result_id),

    CONSTRAINT UQ_RACE_RESULT_registration
        UNIQUE (registration_id),

    CONSTRAINT CK_RACE_RESULT_overall_position
        CHECK
        (
            overall_position IS NULL
            OR overall_position > 0
        ),

    CONSTRAINT CK_RACE_RESULT_category_position
```

CHECK

(

category_position IS NULL

OR category_position > 0

),

CONSTRAINT CK_RACE_RESULT_status

CHECK

(

status IN

('Finished', 'DNF', 'DNS')

),

CONSTRAINT FK_RACE_RESULT_REGISTRATION

FOREIGN KEY (registration_id)

REFERENCES REGISTRATION(registration_id)

);

GO

-- 9. SPONSOR

CREATE TABLE SPONSOR

(

sponsor_id INT IDENTITY(1,1) NOT NULL,

sponsor_name NVARCHAR(100) NOT NULL,

contact_name NVARCHAR(100) NOT NULL,
email NVARCHAR(100) NOT NULL,
phone NVARCHAR(20) NOT NULL,
website NVARCHAR(200) NULL,
sponsorship_level NVARCHAR(20) NOT NULL,

CONSTRAINT PK_SPONSOR

PRIMARY KEY (sponsor_id),

CONSTRAINT UQ_SPONSOR_name

UNIQUE (sponsor_name),

CONSTRAINT UQ_SPONSOR_email

UNIQUE (email),

CONSTRAINT CK_SPONSOR_level

CHECK

(

sponsorship_level IN

('Gold', 'Silver', 'Bronze', 'Other')

)

);

GO

-- 10. RACE_SPONSOR

CREATE TABLE RACE_SPONSOR

(

 race_id INT NOT NULL,

 sponsor_id INT NOT NULL,

 CONSTRAINT PK_RACE_SPONSOR

 PRIMARY KEY (race_id, sponsor_id),

 CONSTRAINT FK_RACE_SPONSOR_RACE

 FOREIGN KEY (race_id)

 REFERENCES RACE(race_id),

 CONSTRAINT FK_RACE_SPONSOR_SPONSOR

 FOREIGN KEY (sponsor_id)

 REFERENCES SPONSOR(sponsor_id)

);

GO

-- INSERT ORGANIZERS

INSERT INTO ORGANIZER

(

 organizer_name,

```
contact_name,
email,
phone,
website
)
VALUES
(
    'RaceDay Events',
    'Thabo Mokoena',
    'thabo@raceday.co.za',
    '0115550101',
    'https://www.raceday.co.za'
),
(
    'City Athletics Club',
    'Lerato Dlamini',
    'lerato@cityathletics.co.za',
    '0115550102',
    'https://www.cityathletics.co.za'
);

GO

-- INSERT PARTICIPANTS / RUNNERS
```

```
INSERT INTO RUNNER
```

```
(  
    first_name,  
    last_name,  
    date_of_birth,  
    gender,  
    email,  
    phone,  
    address,  
    registration_date
```

```
)
```

```
VALUES
```

```
(  
    'Mlungisi',  
    'Makhanya',  
    '2002-05-14',  
    'Male',  
    'mlungisi@example.com',  
    '0825551001',  
    'Johannesburg, Gauteng',  
    '2026-01-15'
```

```
),
```

```
(  
    'Amahle',  
    'Dlamini',  
    '2001-09-22',
```

```
'Female',
'amahle@example.com',
'0825551002',
'Soweto, Gauteng',
'2026-01-18'
),
(
'Kabelo',
'Molefe',
'1995-03-11',
'Male',
'kabelo@example.com',
'0825551003',
'Pretoria, Gauteng',
'2026-02-02'
);
```

GO

-- INSERT LOCATIONS

INSERT INTO LOCATION

```
(
location_name,
address,
```

```
city,
province,
postal_code,
latitude,
longitude
)
VALUES
(
    'Zoo Lake Sports Grounds',
    '57 Lower Park Drive',
    'Johannesburg',
    'Gauteng',
    '2193',
    -26.155300,
    28.026700
),
(
    'Loftus Versfeld Stadium',
    '440 Kirkness Street',
    'Pretoria',
    'Gauteng',
    '0002',
    -25.753300,
    28.226700
),
(
```

```
'Emmarentia Botanical Garden',  
'Olifants Road',  
'Johannesburg',  
'Gauteng',  
'2195',  
-26.155900,  
28.008000  
);
```

```
GO
```

```
-- INSERT 3 EVENTS / RACES
```

```
INSERT INTO RACE
```

```
(  
    race_name,  
    description,  
    race_date,  
    location_id,  
    race_type,  
    entry_fee,  
    capacity,  
    organizer_id,  
    status  
)
```

VALUES

(

'Johannesburg City 10K',

'Annual 10 kilometre road race through Johannesburg.',

'2026-10-10',

1,

'10km',

180.00,

5000,

1,

'Scheduled'

),

(

'Pretoria Spring Half Marathon',

'A 21 kilometre road race celebrating the spring season.',

'2026-10-25',

2,

'21km',

250.00,

4000,

2,

'Scheduled'

),

(

'Emmarentia Family 5K',

'A family-friendly 5 kilometre community fun run.',

```
'2026-11-07',  
3,  
'5km',  
100.00,  
2500,  
1,  
'Scheduled'  
);
```

```
GO
```

```
-- INSERT CATEGORIES FOR EACH EVENT
```

```
INSERT INTO RACE_CATEGORY
```

```
(  
    race_id,  
    category_name,  
    gender_eligibility,  
    age_from,  
    age_to  
)
```

```
VALUES
```

```
-- Event 1
```

```
(  
    1,
```


'Open Male',
'Male',
18,
39
,
(
1,
'Open Female',
'Female',
18,
39
,
(
1,
'Masters Open',
'Open',
40,
99
,

-- Event 2

(
2,
'Senior Male',
'Male',
18,

39

),

(

2,

'Senior Female',

'Female',

18,

39

),

(

2,

'Masters Open',

'Open',

40,

99

),

-- Event 3

(

3,

'Junior Open',

'Open',

13,

17

),

(

```
3,  
'Adult Open',  
'Open',  
18,  
59  
)  
(  
3,  
'Family Open',  
'Open',  
13,  
99  
);
```

```
GO
```

```
-- INSERT SPONSORS
```

```
INSERT INTO SPONSOR
```

```
(  
    sponsor_name,  
    contact_name,  
    email,  
    phone,
```

```
website,  
sponsorship_level  
)  
VALUES  
(  
  'MTN South Africa',  
  'Ayanda Nkosi',  
  'events@mtn.co.za',  
  '0835552001',  
  'https://www.mtn.co.za',  
  'Gold'  
)  
(  
  'Nedbank',  
  'Jason Smith',  
  'sports@nedbank.co.za',  
  '0835552002',  
  'https://www.nedbank.co.za',  
  'Gold'  
)  
(  
  'PUMA South Africa',  
  'Zanele Maseko',  
  'racing@puma.co.za',  
  '0835552003',  
  'https://za.puma.com',
```

```
'Silver'

);

GO

-- =====
-- INSERT RACE-SPONSOR RELATIONSHIPS
-- =====

INSERT INTO RACE_SPONSOR
(
    race_id,
    sponsor_id
)
VALUES
(1, 1),
(1, 2),
(2, 2),
(2, 3),
(3, 1),
(3, 3);

GO

-- =====
-- INSERT SAMPLE ENROLMENTS / REGISTRATIONS
```

-- =====

INSERT INTO REGISTRATION

```
(
    runner_id,
    race_id,
    registration_date,
    status,
    amount_paid,
    payment_status,
    bib_number
```

)

VALUES

```
(
    1,
    1,
    '2026-08-01',
    'Confirmed',
    180.00,
    'Paid',
    1001
```

),

```
(
    2,
    1,
    '2026-08-03',
```

'Confirmed',
180.00,
'Paid',
1002
,
(
3,
2,
'2026-08-05',
'Confirmed',
250.00,
'Paid',
2001
,
(
1,
2,
'2026-08-07',
'Confirmed',
250.00,
'Paid',
2002
,
(
2,
3,

```
'2026-08-10',  
'Confirmed',  
100.00,  
'Paid',  
3001  
);
```

```
GO
```

```
-- =====  
-- INSERT SAMPLE PAYMENTS  
-- =====
```

```
INSERT INTO PAYMENT  
(  
    registration_id,  
    payment_date,  
    amount,  
    payment_method,  
    transaction_reference,  
    status  
)  
VALUES  
(  
    1,  
    '2026-08-01T10:15:00',
```



```
180.00,  
'Card',  
'RD-2026-0001',  
'Success'  
)  
(  
2,  
'2026-08-03T11:20:00',  
180.00,  
'EFT',  
'RD-2026-0002',  
'Success'  
)  
(  
3,  
'2026-08-05T09:45:00',  
250.00,  
'Card',  
'RD-2026-0003',  
'Success'  
)  
(  
4,  
'2026-08-07T14:10:00',  
250.00,  
'EFT',
```

```
'RD-2026-0004',
'Success'
),
(
5,
'2026-08-10T16:30:00',
100.00,
'Card',
'RD-2026-0005',
'Success'
);
```

GO

```
-- =====
```

```
-- INSERT SAMPLE RESULTS
```

```
-- =====
```

```
INSERT INTO RACE_RESULT
```

```
(
  registration_id,
  finish_time,
  overall_position,
  category_position,
  chip_time,
  status
```

```
)  
VALUES  
(  
    1,  
    '00:52:34',  
    1,  
    1,  
    '00:52:10',  
    'Finished'  
)  
(  
    2,  
    '00:58:21',  
    8,  
    4,  
    '00:57:55',  
    'Finished'  
);  
  
GO  
  
-- =====  
-- VERIFY THE DATABASE  
-- =====  
  
SELECT * FROM RUNNER;
```

```
SELECT * FROM ORGANIZER;
```

```
SELECT * FROM LOCATION;
```

```
SELECT * FROM RACE;
```

```
SELECT * FROM RACE_CATEGORY;
```

```
SELECT * FROM REGISTRATION;
```

```
SELECT * FROM PAYMENT;
```

```
SELECT * FROM RACE_RESULT;
```

```
SELECT * FROM SPONSOR;
```

```
SELECT * FROM RACE_SPONSOR;
```

```
GO
```